Live Causal Navigator

Pneumococcal vaccination and pneumonia

Module F · Standalone workshop

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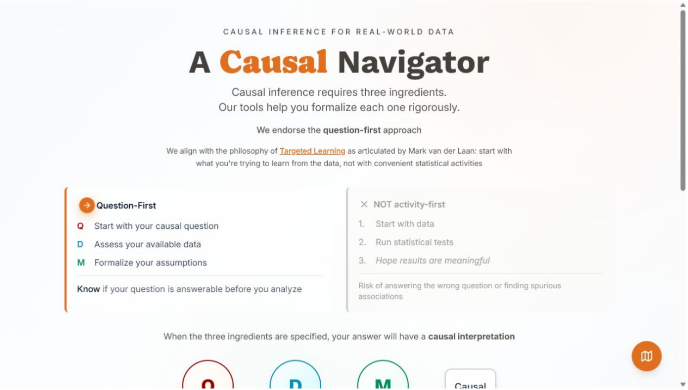
The roadmap becomes a working record

MODULE D → MODULE F

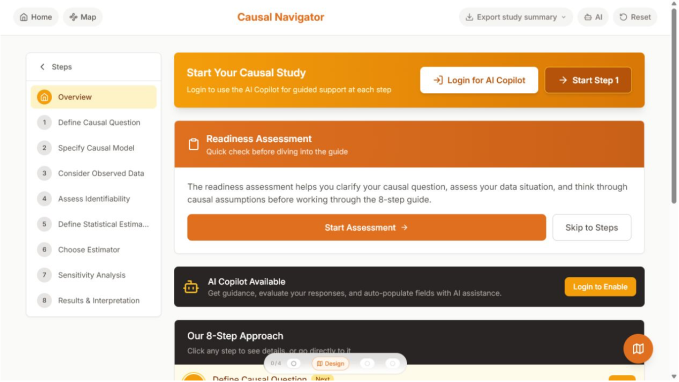
Same causal logic. Now every decision is visible, revisable, and exportable.

Workshop stance

The Navigator is documentation infrastructure—not an oracle.



Eight steps, no login, one export



- **Start at Step 1.** Use the manual route.
- **Work in order.** Pause at design decisions.
- **Export at Step 8.** Keep the Markdown audit trail.

AI support is optional and is not used in this workshop.

navigator.tao-rwd.com

The crude association reflects confounding by indication

Observed 12-month pneumonia risk

7.19%
vaccinated

6.43%
unvaccinated

Crude RD

$+0.76$ **percentage points**

Individual prediction → pair check
Before seeing the prepared answer, name the population, time zero, outcome window, and contrast needed to interpret this difference.

Higher-risk adults are more likely to be vaccinated. The association mixes baseline risk with the treatment effect.

Lock the target trial before the result

Population	Synthetic adult cohort, $N = 5{,}000$; age centered at 65 (not restricted to age 65+)
Treatment	Vaccination at baseline ($A = 1$)
Comparator	No vaccination at baseline ($A = 0$)
Outcome	Pneumonia hospitalization within 12 months
Contrast	Average 12-month risk difference

Time-zero rule

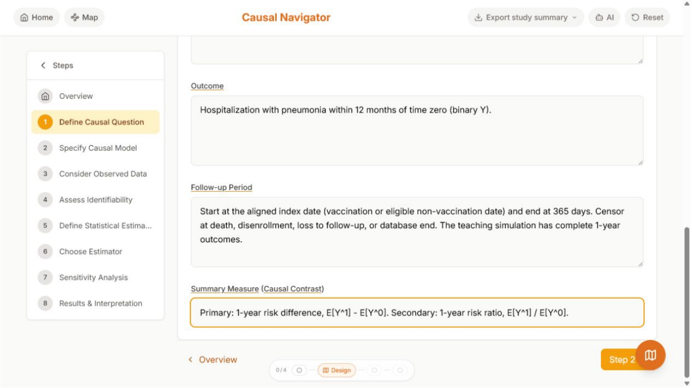
Eligibility, treatment assignment, and follow-up all begin at the same time zero.

Step 1: question-first means fully specified

1 Define the causal question

- intervention, comparator, and time zero
- population and eligibility
- outcome horizon
- causal contrast

Checkpoint
Which single phrase would most change the estimand?



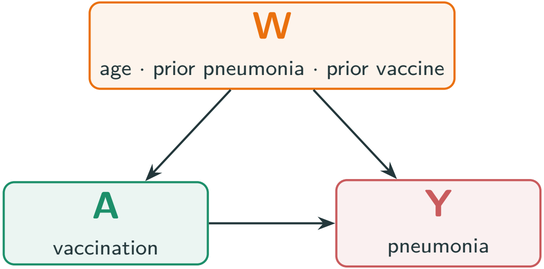
Step 2: the DAG names the adjustment set

THE WORKSHOP VIEW

A compact model makes the confounding structure visible.

- Stipulated baseline set: age, prior pneumonia, and prior vaccination.
- Do not adjust for post-baseline descendants.
- The kit contains the full five-node, nine-edge DAG.

Accept, correct, or reject?
Does this adjustment set follow from the DAG we are willing to defend?



Collapsed discussion view; use the complete DAG for the audit.

Step 3: make the protocol computable

3 Map the data

5,000 people

1,627 vaccinated

334 events

The fixed CSV has one row per person, complete baseline covariates, treatment, and 12-month outcome.

Protocol element	Observed-data mapping
Population	One eligibility-defined row per participant
Treatment	Binary baseline vaccination indicator A
Outcome	Binary 12-month pneumonia hospitalization Y
Baseline W	Age, prior pneumonia, prior vaccination
Time zero	Eligibility, treatment assignment, and follow-up aligned

Mapping check: Can every protocol element be operationalized without changing the question?

Step 4 is the audit gate

Can the observed data answer the causal question?

- exchangeability given W
- practical positivity
- consistency
- no interference

Three-part audit
Choose the weakest assumption. Name the evidence you would seek and one design change if it fails.

Assumption	Data can inform	Still requires judgment
Exchangeability	Covariate balance; proxies	Whether key common causes are measured
Positivity	Fitted score range; support	Whether treatment is possible in every target stratum
Consistency	Recorded treatment versions	Whether the intervention is well defined
No interference	Cluster/context clues	Whether one person's treatment affects another

For this exercise the generator makes the assumptions explicit; the observed data alone still cannot certify them.

Step 5: translate the causal target

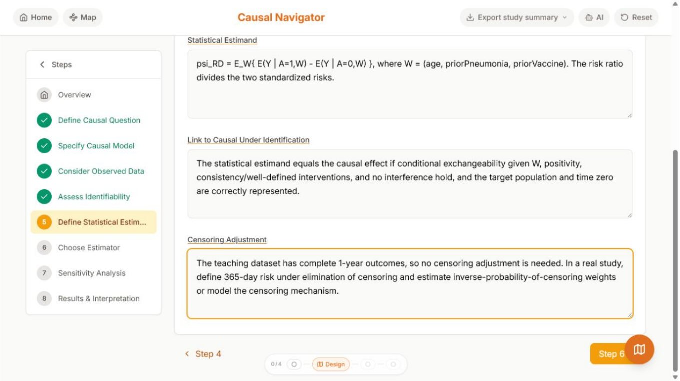
CAUSAL TARGET

$$\psi = E[Y^1] - E[Y^0]$$

OBSERVED-DATA MAPPING

$$E_W\Big[E(Y \mid A=1, W) - E(Y \mid A=0, W)\Big]$$

Audit question
What changes if the population, outcome window, or censoring intervention changes?



Step 6: choose an estimator for that target

6 Choose the estimator

- **Primary:** TMLE for the risk difference
- $\bar{Q}(A, W)$: outcome regression
- $g(W)$: treatment mechanism
- influence-function 95% CI

Boundary question
What can an estimator repair—and what can it never repair?

Outcome-blind work actually completed
No repeated simulation, bias, or RMSE study is included. The adjusted outcome result remains hidden until Steps 1–7 are locked.

Overlap diagnostic
Fitted parametric propensity scores range from 0.0765 to 0.9091. This is a diagnostic—not proof of structural positivity.

Reproducibility convention
Report the version-sensitive TMLE result rounded to -3.0 points (95% CI $-4.6, -1.5$).

Step 7: sensitivity is part of the design

Prespecify the stress tests before the result.

- unmeasured confounding
- outcome misclassification
- selection and censoring
- alternative learners and truncation

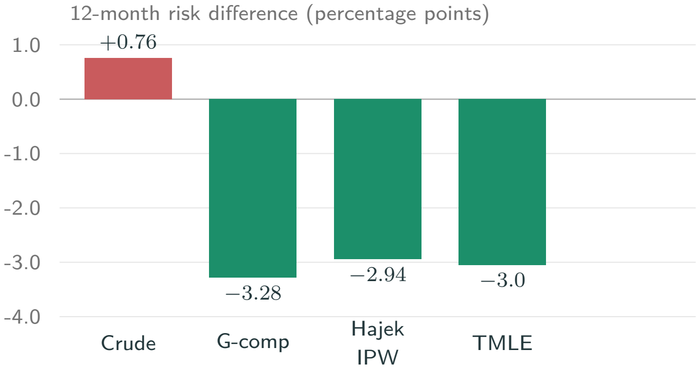
Choose one gap
What analysis or new evidence would make that gap more transparent?

Sampling uncertainty
The influence-function interval quantifies uncertainty for the statistical estimand under the fitted analysis.

Causal uncertainty
Separate analyses must address unmeasured confounding, measurement, selection, treatment versions, and interference.

Descriptive tipping point
A $+3.0$-point shift would move the TMLE point estimate to the null. This is not a formal E-value.

Adjustment reverses the sign



Synthetic teaching result — not clinical evidence.

Crude
+0.76 pp

TMLE
−3.0 pp

95% CI: −4.6 to −1.5 pp

What this shows

Adjusted analyses tell a different story under the stipulated model. Agreement does not prove the assumptions.

Step 8: interpretation inherits every prior assumption

DEFENSIBLE

"In this synthetic exercise, adjustment reverses the crude association under the stipulated causal model."

Supported claim

A statement about this workflow, this fixed dataset, and the assumptions explicitly recorded in the roadmap.

NOT DEFENSIBLE

"The vaccine prevents pneumonia in real patients."

Still unknown

Clinical effectiveness, safety, transportability, real EHR bias, and population transmission effects.

Accept, correct, or reject?

Cite one Step 4 assumption and one limitation when revising the conclusion.

YOUR DECISION LOG

Choose one field from Steps 1–8.

1. What evidence makes the entry defensible?
2. What would make you revise it?
3. Which downstream decision changes if you revise it?

Pair protocol

Accept, correct, or reject the entry. Compare reasoning before export.

Participant kit

Participant workbook PDF
Complete DAGitty specification
Fixed synthetic dataset
Reproducible analysis script

Answer key—after the activity

Completed eight-step roadmap in Markdown

The estimator is downstream of the science

Module F leaves us with:

- a specified target trial
- an explicit causal model
- an identification argument
- a statistical estimand
- a prespecified estimator and sensitivity plan

MODULE G HANDOFF

Use the same cohort and estimand to compare adjusted estimates, uncertainty, and the limits that remain.

Design first

Estimate second. Interpret last.